CS383 · SOFTWARE ENGINEERING

Mock Midterm

A practice exam in the midterm's format · 60 minutes · 20 marks

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01 Before you start

This exam is my own, in the format of Model 1 (the newest midterm): 20 marks and five questions. The texts and numbers are new.

Question	1	2	3	4	5	Total
Marks	2	4	4	4	6	20

- Set a 60-minute timer.
- The solutions are in a separate file. Open it only after you finish.

02 Questions

Q1 MCQ · 2 PTS

Multiple choice questions

1. Which of the following is a customized software product?

☐ Microsoft Excel

☐ A payroll system built for one specific company

☐ A CAD package

☐ A web browser

2. One benefit of Software as a Service (SaaS) in the slides is:

☐ Full control

☐ Pay as you go

☐ No internet needed

☐ Unlimited range of applications

3. The outputs of sprint planning are:

☐ Team capacity and product backlog

☐ Sprint goal and sprint backlog

☐ Business conditions and technology

☐ Velocity and focus factor

4. Which of the following is NOT one of the four factors a project manager considers in people management?

☐ Consistency ☐ Respect ☐ Speed ☐ Honesty

Q2 ETHICS · 4 PTS

According to the 8 principles of the IEEE Code of Ethics (PUBLIC, CLIENT AND EMPLOYER, PRODUCT, JUDGMENT, MANAGEMENT, PROFESSION, COLLEAGUES, SELF), identify the type of ethics reflected in the following descriptions:

SEEPP ethic example	Ethics Principle
Review the work of others in an objective, candid, and properly-documented way.	
Identify, document, and report significant issues of social concern, of which they are aware, in software or related documents, to the employer or the client.	
Ensure that software engineers know the employer's policies and procedures for protecting passwords, files and information that is confidential.	
Not promote their own interest at the expense of the profession, client or employer.	

Q3 SPRINT CALCULATIONS · 4 PTS

Team of 4 people; workday = 8h. Last 5 sprints completed story points: 20, 26, 24, 22, 28.

1. Compute the sprint velocity. (1pts)
2. Compute the focus factor. (1pts)
3. For a 2-week sprint (Mon–Fri, weekends off), compute the effective team capacity (hours). (1pts)
4. If the next sprint backlog is 26 story points, state whether the plan is feasible and why. (1pts)

Q4 PROCESS MODEL · 4 PTS

For each project, choose one development process model: Agile, Plan-Driven, or Component-Based.

Project 1 — Online Store. Requirements are moderately stable. The store must integrate an existing payment gateway, a shipping company's API and an off-the-shelf product catalogue component. 10 engineers.

Project 2 — Railway Signalling Control. Safety-critical. The requirements are fixed by the railway authority and reviewed only at stage gates. 35 engineers in several departments. Full documentation and sign-offs are required.

Project 3 — Student Club Events App. The club's president meets the team every day. The features change every week. 4 co-located developers. A first version is needed in three weeks.

Project 4 — Government Tax Reporting. Report templates are fixed by regulation for five years. Auditors set the requirements up front. 25 people with a formal project office. High penalty for defects.

Q5 PRODUCT / ORGANIZATION / EXTERNAL · 6 PTS

For each statement, indicate the requirement type: Product, Organization or External.

Statement	Type
1. The system shall encrypt all stored passwords.	
2. The website must meet the accessibility standard required by law.	
3. The system must run on the company's existing database servers.	
4. The home page shall load in under 2 seconds on a 4G connection.	
5. Developers must document every module using the organization's documentation template.	
6. Financial records must be kept for 10 years as required by the tax authority.	